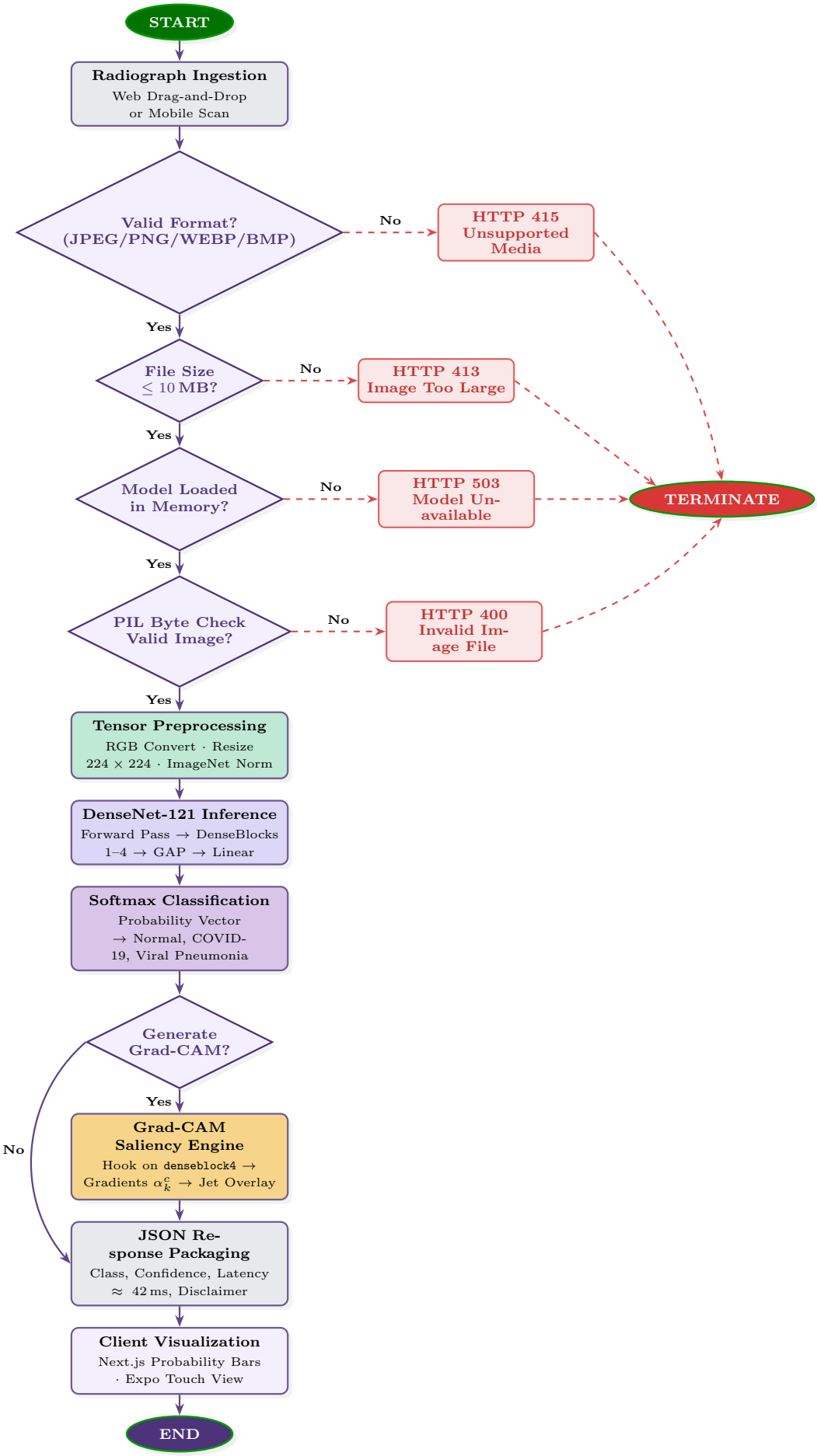

START
Radiograph Ingestion
Web Drag-and-Drop
or Mobile Scan
Valid Format?
(JPEG/PNG/WEBP/BMP)
No
HTTP 415
Unsupported
Media
Yes
File Size
≤ 10 MB?
No
HTTP 413
Image Too Large
Yes
Model Loaded
in Memory?
No
HTTP 503
Model Un-
available
TERMINATE
Yes
PIL Byte Check
Valid Image?
No
HTTP 400
Invalid Im-
age File
Yes
Tensor Preprocessing
RGB Convert · Resize
224 × 224 · ImageNet Norm
DenseNet-121 Inference
Forward Pass → DenseBlocks
1–4 → GAP → Linear
Softmax Classification
Probability Vector
→ Normal, COVID-
19, Viral Pneumonia
Generate
Grad-CAM?
Yes
No
Grad-CAM
Saliency Engine
Hook on denseblock4 →
Gradients α_k^c → Jet Overlay
JSON Re-
sponse Packaging
Class, Confidence, Latency
≈ 42 ms, Disclaimer
Client Visualization
Next.js Probability Bars
· Expo Touch View
END